ANX-PR/CL/001-01
LEARNING GUIDE

SUBJECT

103000856 - Deep Learning

DEGREE PROGRAMME

10AZ - Master Universitario En Innovación Digital

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

Index

Learning guide

1. Description

1.1. Subject details

Name of the subject	103000856 - Deep Learning
No of credits	3 ECTS
Type	Optional/elective
Academic year ot the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	10AZ - Master Universitario en Innovación Digital
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Luis Baumela Molina	2207	luis.baumela@upm.es	Sin horario. Appointment required via email.
Roberto Valle Fernandez (Subject coordinator)	2101	roberto.valle@upm.es	Sin horario. Appointment required via email.

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Computer languages (e.g., Python)

- Basic foundations of artificial neural networks (e.g., shallow neural networks and backpropagation algorithm).

4. Skills and learning outcomes *

4.1. Skills to be learned

CB07 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio

CE-CD08 - Capacidad para utilizar y seleccionar las herramientas más adecuadas para deep learning

CG03 - La capacidad de usar la lengua inglesa de manera competente, es decir, con capacitación para tareas complejas de trabajo y estudio.

4.2. Learning outcomes

RA64 - Apply machine learning software tools for practical problems related to deep learning

RA63 - Identify areas of application where deep learning techniques can be used

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

The course presents a theoretical and practical view of deep learning. It is assumed that students are familiar with the basic foundations of shallow neural networks. The course provides an introduction to deep neural networks, covering key training methods such as backpropagation, optimization algorithms, and regularization techniques. It then focuses on deep learning applications in computer vision, including convolutional neural networks (CNNs), state-of-the-art architectures, representation learning, and tasks such as object detection and image segmentation.

5.2. Syllabus

1. Introduction to deep neural networks
   1.1. Backpropagation algorithm
   1.2. Optimization algorithms
   1.3. Regularization
2. Deep learning for computer vision
   2.1. Foundations of computer vision
   2.2. Convolutional neural networks
   2.3. Popular architectures
   2.4. Representation learning
   2.5. Applications

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	**Lecture on Unit 1** Duration: 02:00 Lecture **Lecture on Unit 1** Duration: 02:00 Lecture			
2	**Lecture on Unit 1** Duration: 02:00 Lecture **Lecture on Unit 1** Duration: 02:00 Lecture			
3	**Lecture on Unit 1** Duration: 02:00 Lecture **Lecture on Unit 1** Duration: 02:00 Lecture			
4	**Assessment activity for Unit 1** Duration: 02:00 Additional activities **Lecture on Unit 2** Duration: 02:00 Lecture			**Assessment activity for Unit 1** Written test Progressive assessment Presential Duration: 02:00
5	**Lecture on Unit 2** Duration: 02:00 Lecture **Lecture on Unit 2** Duration: 02:00 Lecture			
6	**Lecture on Unit 2** Duration: 02:00 Lecture **Lecture on Unit 2** Duration: 02:00 Lecture			

7	**Lecture on Unit 2** Duration: 02:00 Lecture **Lecture on Unit 2** Duration: 02:00 Lecture			
8	**Assessment activity for Unit 2** Duration: 02:00 Additional activities			**Assessment activity for Unit 2** Written test Progressive assessment Presential Duration: 02:00
9				
10				
11				
12				
13				
14				
15				
16				
17				**Practical assignment** Group work Progressive assessment and Global Examination Not Presential Duration: 00:00 **Assessment activity for Unit 1** Written test Global examination Presential Duration: 02:00 **Assessment activity for Unit 2** Written test Global examination Presential Duration: 02:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
4	Assessment activity for Unit 1	Written test	Face-to-face	02:00	30%	4 / 10	CB07 CG03 CE-CD08
8	Assessment activity for Unit 2	Written test	Face-to-face	02:00	30%	4 / 10	CB07 CG03 CE-CD08
17	Practical assignment	Group work	No Presential	00:00	40%	4 / 10	CB07 CG03 CE-CD08

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Practical assignment	Group work	No Presential	00:00	40%	4 / 10	CB07 CG03 CE-CD08
17	Assessment activity for Unit 1	Written test	Face-to-face	02:00	30%	4 / 10	CB07 CG03 CE-CD08
17	Assessment activity for Unit 2	Written test	Face-to-face	02:00	30%	4 / 10	CB07 CG03 CE-CD08

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Assessment activity for Unit 1 and Unit 2	Written test	Face-to-face	02:00	60%	4 / 10	CB07 CG03 CE-CD08

Practical assignment	Group work	Face-to-face	00:00	40%	4 / 10	CB07 CG03 CE-CD08

7.2. Assessment criteria

Under the progressive evaluation system, there are 3 graded components:

1. Exam on Unit 1. This exam will take place around week 4. Students who do not reach the minimum required score must retake the exam during the regular exam session. Students who meet the minimum score but receive a grade below 5 out of 10 may also choose to retake the exam during the regular session. In that case, the final grade will be the one obtained in the second attempt. Students who score above 5 may not retake this exam during the regular session.
2. Exam on Unit 2. This exam will take place around week 8. Students who do not reach the minimum required score must retake the exam during the regular exam session. Students who meet the minimum score but receive a grade below 5 out of 10 may also choose to retake the exam during the regular session. In that case, the final grade will be the one obtained in the second attempt. Students who score above 5 may not retake this exam during the regular session
3. Practical project. Throughout the course, as new content is introduced, students will work in groups to develop a practical computer vision project. This includes developing a software application and writing a report. The final submission of the project coincides with the regular exam session. Students who receive a grade above 5 for the project may not resubmit it during the resit (extraordinary) session.

The final grade is a weighted sum of the three assessment components listed above. To pass the course, students must meet the minimum required score in each component and achieve an overall weighted average of 5 or higher out of 10. If a student fails to meet the minimum score in any component, their final course grade will be a maximum of 4 out of 10 (fail).

In accordance with UPM regulations, exam solutions will be published within two working days after the exam. To avoid redundancy, theoretical answers that are direct transcriptions from the course materials may be omitted from the solution key, as these materials are available on the Moodle platform.

In the extraordinary session, an exam covering all course content will be administered. Students who did not meet the minimum required grade for the practical project during the regular session must resubmit the project for the extraordinary session.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
UPM Moodle	Web resource	All course materials will be available on the university's Moodle platform.
Bibliography	Bibliography	Selected bibliography (papers and text books).

9. Other information

9.1. Other information about the subject

The course "Deep Learning" is related to the "Sustainable Development Goal 9" (Build resilient infrastructure, promote sustainable industrialization and foster innovation), defined by the United Nations Development Programme (www.undp.org) in terms of innovation and scientific research in information technologies.

The information contained in this learning guide is for guidance only and may vary due to errors, omissions, changes in the health situation, changes in applicable regulations, or incidents that occur throughout the semester.